In [6]:
```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings("ignore")
```

In [7]:
```
from sklearn.datasets import load_iris
iris=load_iris()
iris
```

Out[7]:
```
{'data': array([[5.1, 3.5, 1.4, 0.2],
                [4.9, 3. , 1.4, 0.2],
                [4.7, 3.2, 1.3, 0.2],
                [4.6, 3.1, 1.5, 0.2],
                [5. , 3.6, 1.4, 0.2],
                [5.4, 3.9, 1.7, 0.4],
                [4.6, 3.4, 1.4, 0.3],
                [5. , 3.4, 1.5, 0.2],
                [4.4, 2.9, 1.4, 0.2],
                [4.9, 3.1, 1.5, 0.1],
                [5.4, 3.7, 1.5, 0.2],
                [4.8, 3.4, 1.6, 0.2],
                [4.8, 3. , 1.4, 0.1],
                [4.3, 3. , 1.1, 0.1],
                [5.8, 4. , 1.2, 0.2],
                [5.7, 4.4, 1.5, 0.4],
                [5.4, 3.9, 1.3, 0.4],
                [5.1, 3.5, 1.4, 0.3],
                [5.7, 3.8, 1.7, 0.3],
                [5.1, 3.8, 1.5, 0.3],
                [5.4, 3.4, 1.7, 0.2],
                [5.1, 3.7, 1.5, 0.4],
                [4.6, 3.6, 1. , 0.2],
                [5.1, 3.3, 1.7, 0.5],
                [4.8, 3.4, 1.9, 0.2],
                [5. , 3. , 1.6, 0.2],
                [5. , 3.4, 1.6, 0.4],
                [5.2, 3.5, 1.5, 0.2],
                [5.2, 3.4, 1.4, 0.2],
                [4.7, 3.2, 1.6, 0.2],
                [4.8, 3.1, 1.6, 0.2],
                [5.4, 3.4, 1.5, 0.4],
                [5.2, 4.1, 1.5, 0.1],
                [5.5, 4.2, 1.4, 0.2],
                [4.9, 3.1, 1.5, 0.2],
                [5. , 3.2, 1.2, 0.2],
                [5.5, 3.5, 1.3, 0.2],
                [4.9, 3.6, 1.4, 0.1],
                [4.4, 3. , 1.3, 0.2],
                [5.1, 3.4, 1.5, 0.2],
```

```
[5. , 3.5, 1.3, 0.3],
[4.5, 2.3, 1.3, 0.3],
[4.4, 3.2, 1.3, 0.2],
[5. , 3.5, 1.6, 0.6],
[5.1, 3.8, 1.9, 0.4],
[4.8, 3. , 1.4, 0.3],
[5.1, 3.8, 1.6, 0.2],
[4.6, 3.2, 1.4, 0.2],
[5.3, 3.7, 1.5, 0.2],
[5. , 3.3, 1.4, 0.2],
[7. , 3.2, 4.7, 1.4],
[6.4, 3.2, 4.5, 1.5],
[6.9, 3.1, 4.9, 1.5],
[5.5, 2.3, 4. , 1.3],
[6.5, 2.8, 4.6, 1.5],
[5.7, 2.8, 4.5, 1.3],
[6.3, 3.3, 4.7, 1.6],
[4.9, 2.4, 3.3, 1. ],
[6.6, 2.9, 4.6, 1.3],
[5.2, 2.7, 3.9, 1.4],
[5. , 2. , 3.5, 1. ],
[5.9, 3. , 4.2, 1.5],
[6. , 2.2, 4. , 1. ],
[6.1, 2.9, 4.7, 1.4],
[5.6, 2.9, 3.6, 1.3],
[6.7, 3.1, 4.4, 1.4],
[5.6, 3. , 4.5, 1.5],
[5.8, 2.7, 4.1, 1. ],
[6.2, 2.2, 4.5, 1.5],
[5.6, 2.5, 3.9, 1.1],
[5.9, 3.2, 4.8, 1.8],
[6.1, 2.8, 4. , 1.3],
[6.3, 2.5, 4.9, 1.5],
[6.1, 2.8, 4.7, 1.2],
[6.4, 2.9, 4.3, 1.3],
[6.6, 3. , 4.4, 1.4],
[6.8, 2.8, 4.8, 1.4],
[6.7, 3. , 5. , 1.7],
[6. , 2.9, 4.5, 1.5],
[5.7, 2.6, 3.5, 1. ],
[5.5, 2.4, 3.8, 1.1],
```

```
[5.5, 2.4, 3.7, 1. ],
[5.8, 2.7, 3.9, 1.2],
[6. , 2.7, 5.1, 1.6],
[5.4, 3. , 4.5, 1.5],
[6. , 3.4, 4.5, 1.6],
[6.7, 3.1, 4.7, 1.5],
[6.3, 2.3, 4.4, 1.3],
[5.6, 3. , 4.1, 1.3],
[5.5, 2.5, 4. , 1.3],
[5.5, 2.6, 4.4, 1.2],
[6.1, 3. , 4.6, 1.4],
[5.8, 2.6, 4. , 1.2],
[5. , 2.3, 3.3, 1. ],
[5.6, 2.7, 4.2, 1.3],
[5.7, 3. , 4.2, 1.2],
[5.7, 2.9, 4.2, 1.3],
[6.2, 2.9, 4.3, 1.3],
[5.1, 2.5, 3. , 1.1],
[5.7, 2.8, 4.1, 1.3],
[6.3, 3.3, 6. , 2.5],
[5.8, 2.7, 5.1, 1.9],
[7.1, 3. , 5.9, 2.1],
[6.3, 2.9, 5.6, 1.8],
[6.5, 3. , 5.8, 2.2],
[7.6, 3. , 6.6, 2.1],
[4.9, 2.5, 4.5, 1.7],
[7.3, 2.9, 6.3, 1.8],
[6.7, 2.5, 5.8, 1.8],
[7.2, 3.6, 6.1, 2.5],
[6.5, 3.2, 5.1, 2. ],
[6.4, 2.7, 5.3, 1.9],
[6.8, 3. , 5.5, 2.1],
[5.7, 2.5, 5. , 2. ],
[5.8, 2.8, 5.1, 2.4],
[6.4, 3.2, 5.3, 2.3],
[6.5, 3. , 5.5, 1.8],
[7.7, 3.8, 6.7, 2.2],
[7.7, 2.6, 6.9, 2.3],
[6. , 2.2, 5. , 1.5],
[6.9, 3.2, 5.7, 2.3],
[5.6, 2.8, 4.9, 2. ],
```

```
        [7.7, 2.8, 6.7, 2. ],
        [6.3, 2.7, 4.9, 1.8],
        [6.7, 3.3, 5.7, 2.1],
        [7.2, 3.2, 6. , 1.8],
        [6.2, 2.8, 4.8, 1.8],
        [6.1, 3. , 4.9, 1.8],
        [6.4, 2.8, 5.6, 2.1],
        [7.2, 3. , 5.8, 1.6],
        [7.4, 2.8, 6.1, 1.9],
        [7.9, 3.8, 6.4, 2. ],
        [6.4, 2.8, 5.6, 2.2],
        [6.3, 2.8, 5.1, 1.5],
        [6.1, 2.6, 5.6, 1.4],
        [7.7, 3. , 6.1, 2.3],
        [6.3, 3.4, 5.6, 2.4],
        [6.4, 3.1, 5.5, 1.8],
        [6. , 3. , 4.8, 1.8],
        [6.9, 3.1, 5.4, 2.1],
        [6.7, 3.1, 5.6, 2.4],
        [6.9, 3.1, 5.1, 2.3],
        [5.8, 2.7, 5.1, 1.9],
        [6.8, 3.2, 5.9, 2.3],
        [6.7, 3.3, 5.7, 2.5],
        [6.7, 3. , 5.2, 2.3],
        [6.3, 2.5, 5. , 1.9],
        [6.5, 3. , 5.2, 2. ],
        [6.2, 3.4, 5.4, 2.3],
        [5.9, 3. , 5.1, 1.8]]),
 'target': array([0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
        0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
        0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
        1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
        1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2,
        2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2,
        2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2]),
 'frame': None,
 'target_names': array(['setosa', 'versicolor', 'virginica'], dtype='<U10'),
 'DESCR': '.. _iris_dataset:\n\nIris plants dataset\n--------------------\n\n**Data Set Characteristics:**\n\n:Number of Inst
ances: 150 (50 in each of three classes)\n:Number of Attributes: 4 numeric, predictive attributes and the class\n:Attribute I
nformation:\n    - sepal length in cm\n    - sepal width in cm\n    - petal length in cm\n    - petal width in cm\n    - clas
s:\n            - Iris-Setosa\n            - Iris-Versicolour\n            - Iris-Virginica\n\n:Summary Statistics:\n\n======
```

```
======== ==== ==== ======= ===== ====================\n                Min  Max   Mean    SD   Class Correlation\n===========
=== ==== ==== ======= ===== ====================\nsepal length:   4.3  7.9   5.84   0.83    0.7826\nsepal width:    2.0  4.4
3.05   0.43   -0.4194\npetal length:   1.0  6.9   3.76   1.76    0.9490  (high!)\npetal width:    0.1  2.5   1.20   0.76
0.9565  (high!)\n============== ==== ==== ======= ===== ====================\n\n:Missing Attribute Values: None\n:Class Distr
ibution: 33.3% for each of 3 classes.\n:Creator: R.A. Fisher\n:Donor: Michael Marshall (MARSHALL%PLU@io.arc.nasa.gov)\n:Date:
July, 1988\n\nThe famous Iris database, first used by Sir R.A. Fisher. The dataset is taken\nfrom Fisher\'s paper. Note that
it\'s the same as in R, but not as in the UCI\nMachine Learning Repository, which has two wrong data points.\n\nThis is perha
ps the best known database to be found in the\npattern recognition literature.  Fisher\'s paper is a classic in the field and
\nis referenced frequently to this day.  (See Duda & Hart, for example.)  The\ndata set contains 3 classes of 50 instances ea
ch, where each class refers to a\ntype of iris plant.  One class is linearly separable from the other 2; the\nlatter are NOT
linearly separable from each other.\n\n.. dropdown:: References\n\n  - Fisher, R.A. "The use of multiple measurements in taxo
nomic problems"\n    Annual Eugenics, 7, Part II, 179-188 (1936); also in "Contributions to\n    Mathematical Statistics" (Jo
hn Wiley, NY, 1950).\n  - Duda, R.O., & Hart, P.E. (1973) Pattern Classification and Scene Analysis.\n    (Q327.D83) John Wil
ey & Sons.  ISBN 0-471-22361-1.  See page 218.\n  - Dasarathy, B.V. (1980) "Nosing Around the Neighborhood: A New System\n
Structure and Classification Rule for Recognition in Partially Exposed\n    Environments".  IEEE Transactions on Pattern Anal
ysis and Machine\n    Intelligence, Vol. PAMI-2, No. 1, 67-71.\n  - Gates, G.W. (1972) "The Reduced Nearest Neighbor Rule".
IEEE Transactions\n    on Information Theory, May 1972, 431-433.\n  - See also: 1988 MLC Proceedings, 54-64.  Cheeseman et a
l"s AUTOCLASS II\n    conceptual clustering system finds 3 classes in the data.\n  - Many, many more ...\n',
 'feature_names': ['sepal length (cm)',
  'sepal width (cm)',
  'petal length (cm)',
  'petal width (cm)'],
 'filename': 'iris.csv',
 'data_module': 'sklearn.datasets.data'}
```

In [8]:

```
df=pd.DataFrame(iris.data,columns=iris.feature_names)
df.head()
df["Class"]=iris.target
df.head()
```

Out[8]:

| | sepal length (cm) | sepal width (cm) | petal length (cm) | petal width (cm) | Class |
|---|---|---|---|---|---|
| **0** | 5.1 | 3.5 | 1.4 | 0.2 | 0 |
| **1** | 4.9 | 3.0 | 1.4 | 0.2 | 0 |
| **2** | 4.7 | 3.2 | 1.3 | 0.2 | 0 |
| **3** | 4.6 | 3.1 | 1.5 | 0.2 | 0 |
| **4** | 5.0 | 3.6 | 1.4 | 0.2 | 0 |

In [9]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   sepal length (cm)  150 non-null    float64
 1   sepal width (cm)   150 non-null    float64
 2   petal length (cm)  150 non-null    float64
 3   petal width (cm)   150 non-null    float64
 4   Class              150 non-null    int32
dtypes: float64(4), int32(1)
memory usage: 5.4 KB
```

In [10]:
```
df[df.duplicated()]
```

Out[10]:

| | sepal length (cm) | sepal width (cm) | petal length (cm) | petal width (cm) | Class |
|---|---|---|---|---|---|
| **142** | 5.8 | 2.7 | 5.1 | 1.9 | 2 |

In [11]:
```
# target - class :- categorical ---classification
```

In [12]:
```
# exploring target
df["Class"].value_counts()
# balanced datset
```

Out[12]:
```
Class
0    50
1    50
2    50
Name: count, dtype: int64
```

In [13]:
```
# statistical distribution check
df.describe()
# +ve skeweness - sepal length (cm)     a little skewess
# -ve skeweness - petal length (cm)     petal width (cm)
```

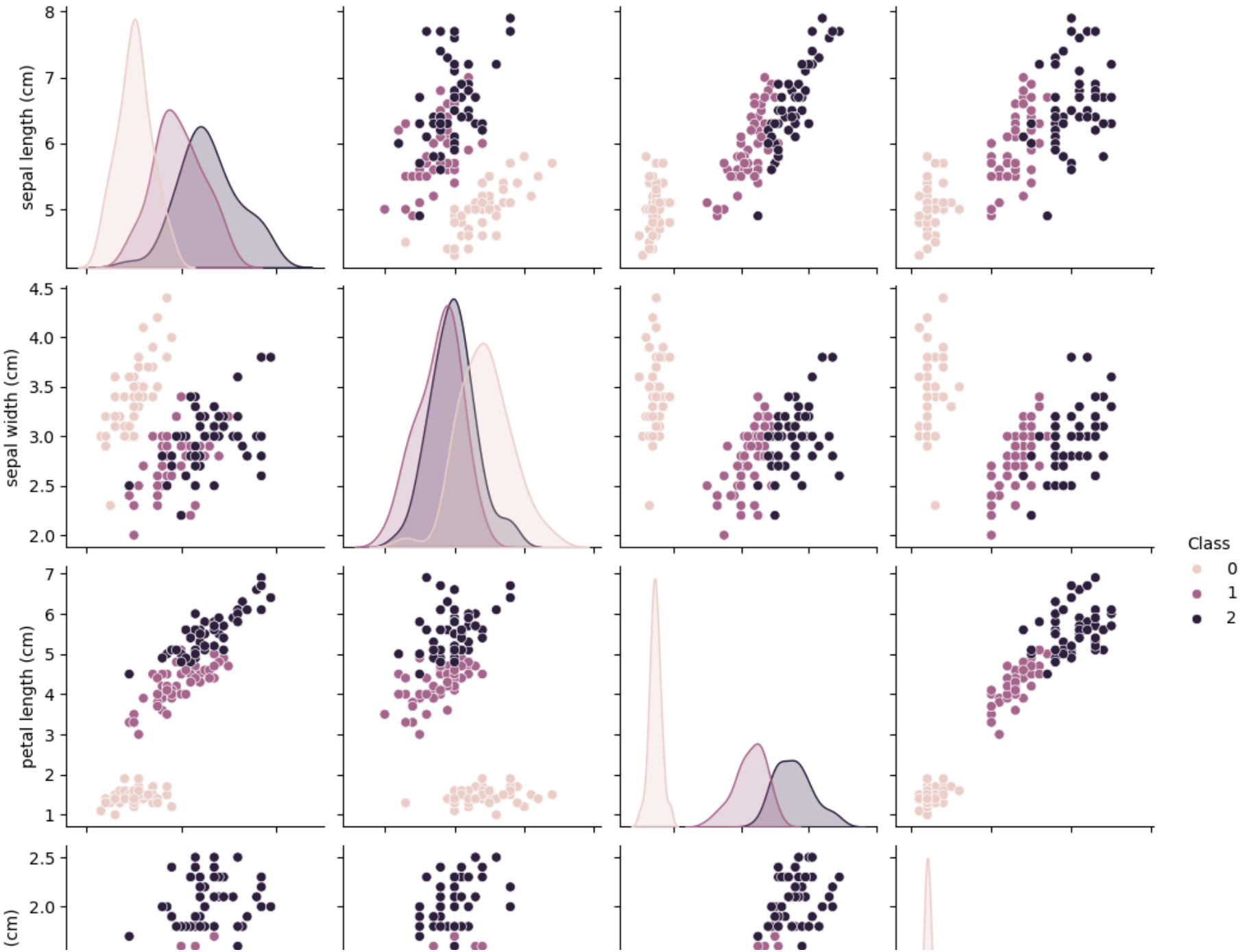
Out[13]:

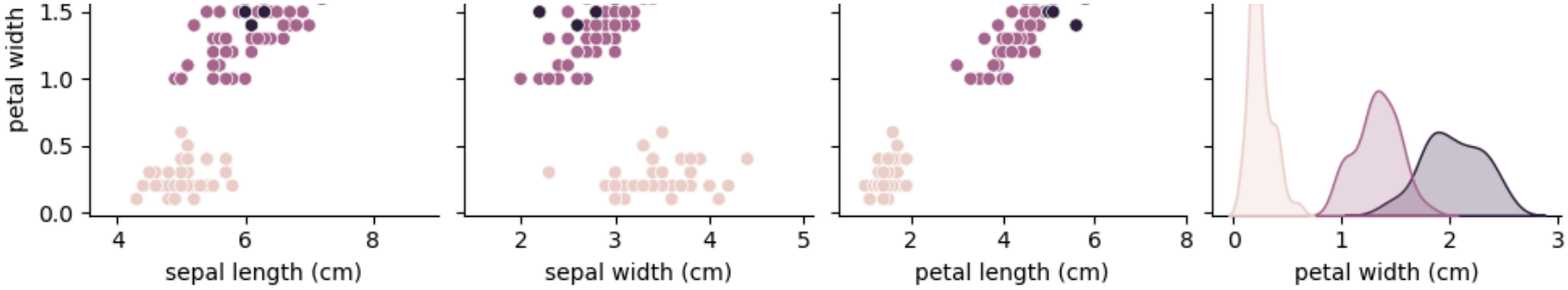
| | sepal length (cm) | sepal width (cm) | petal length (cm) | petal width (cm) | Class |
|---|---|---|---|---|---|
| **count** | 150.000000 | 150.000000 | 150.000000 | 150.000000 | 150.000000 |
| **mean** | 5.843333 | 3.057333 | 3.758000 | 1.199333 | 1.000000 |
| **std** | 0.828066 | 0.435866 | 1.765298 | 0.762238 | 0.819232 |
| **min** | 4.300000 | 2.000000 | 1.000000 | 0.100000 | 0.000000 |
| **25%** | 5.100000 | 2.800000 | 1.600000 | 0.300000 | 0.000000 |
| **50%** | 5.800000 | 3.000000 | 4.350000 | 1.300000 | 1.000000 |
| **75%** | 6.400000 | 3.300000 | 5.100000 | 1.800000 | 2.000000 |
| **max** | 7.900000 | 4.400000 | 6.900000 | 2.500000 | 2.000000 |

In [14]:
```
# skeweness ,boxplot
```

In [15]:
```
sns.pairplot(df,hue="Class")
```

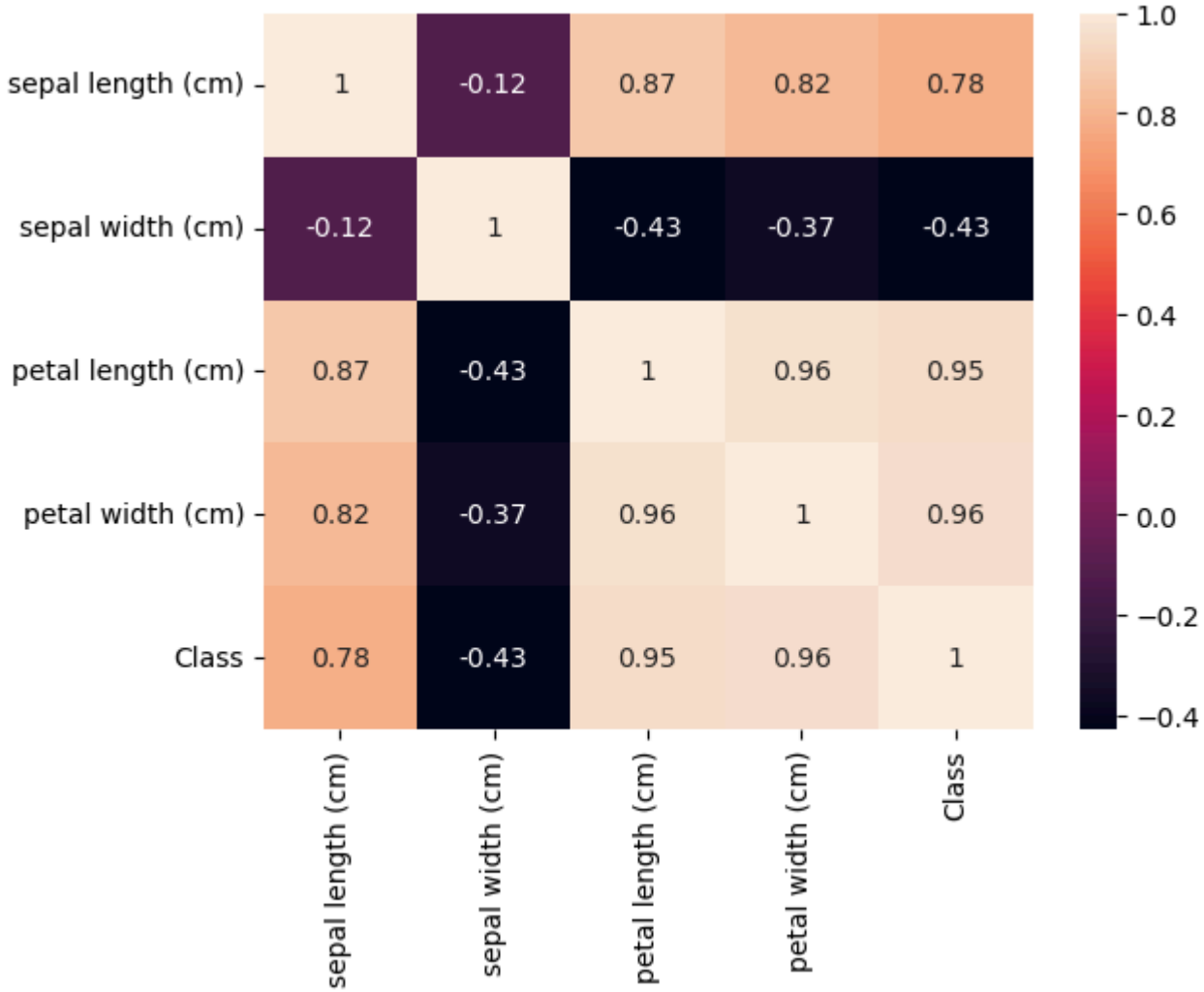
Out[15]:
```
<seaborn.axisgrid.PairGrid at 0x20979473b90>
```


sepal length (cm)
sepal width (cm)
petal length (cm)
(cm)
Class
0
1
2




In [16]:
```
sns.heatmap(df.corr(),annot=True)
```

Out[16]: <Axes: >




In [17]:
```
features=df.drop("Class",axis=1)
target=df["Class"]
```

In [18]:
```
from sklearn.model_selection import train_test_split
xtrain, xtest, ytrain , ytest = train_test_split(features ,target , test_size=0.20,random_state =2)
```

In [19]:
```python
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import classification_report,confusion_matrix
```

In [20]:
```python
def mymodel(model):
    model.fit(xtrain,ytrain)
    ypred=model.predict(xtest)
    # checking for overfit and underfit
    print(f"Training Score : {model.score(xtrain,ytrain)}")
    print(f"Testing score : {model.score(xtest,ytest)}")
    # evalution metrics
    c=pd.DataFrame(confusion_matrix(ytest,ypred),
                   index=["Setosa","Versicolor","Virginica"],
                    columns=["Setosa","Versicolor","Virginica"])
    print(f"Confusion Matric:\n{c}")
    print(f"Classification report :\n{classification_report(ytest,ypred)}")
    return model
```

In [21]:
```python
mymodel(DecisionTreeClassifier())
```

```
Training Score : 1.0
Testing score : 0.9333333333333333
Confusion Matric:
            Setosa  Versicolor  Virginica
Setosa          14           0          0
Versicolor       0           7          1
Virginica        0           1          7
Classification report :
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        14
           1       0.88      0.88      0.88         8
           2       0.88      0.88      0.88         8

    accuracy                           0.93        30
   macro avg       0.92      0.92      0.92        30
weighted avg       0.93      0.93      0.93        30
```

Out[21]:

```
DecisionTreeClassifier
DecisionTreeClassifier()
```

In [22]:

```
dt=mymodel(DecisionTreeClassifier())
dt
```

```
Training Score : 1.0
Testing score : 0.9333333333333333
Confusion Matric:
            Setosa  Versicolor  Virginica
Setosa          14           0          0
Versicolor       0           7          1
Virginica        0           1          7
Classification report :
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        14
           1       0.88      0.88      0.88         8
           2       0.88      0.88      0.88         8

    accuracy                           0.93        30
   macro avg       0.92      0.92      0.92        30
weighted avg       0.93      0.93      0.93        30
```

Out[22]:

```
DecisionTreeClassifier
DecisionTreeClassifier()
```

In [23]:

```
d={"Features":iris.feature_names,
   "Importance":dt.feature_importances_
  }
f=pd.DataFrame(d)
f
```

Out[23]:

| | Features | Importance |
|---|---|---|
| **0** | sepal length (cm) | 0.000000 |
| **1** | sepal width (cm) | 0.018797 |
| **2** | petal length (cm) | 0.056904 |
| **3** | petal width (cm) | 0.924299 |

In [24]:

```
from sklearn import tree
tree.plot_tree(dt,feature_names=iris.feature_names,filled=True)
plt.show()
```

In [25]:

```
dt.get_depth()
```

Out[25]: 4

In [26]:
```
dt.get_n_leaves()
```

Out[26]: 7

In [27]:
```
# to deal with overfitting apply pruning techniques
parameters={
    "max_depth":list(range(2,3)),
    "min_samples_split":list(range(2,4)),
    "min_samples_split":list(range(2,4))
}
```

In [28]:
```
from sklearn.model_selection import GridSearchCV
clf=GridSearchCV(DecisionTreeClassifier(),parameters,verbose=2)
clf.fit(xtrain,ytrain)
```

```
Fitting 5 folds for each of 2 candidates, totalling 10 fits
[CV] END ...................max_depth=2, min_samples_split=2; total time=   0.0s
[CV] END ...................max_depth=2, min_samples_split=2; total time=   0.0s
[CV] END ...................max_depth=2, min_samples_split=2; total time=   0.0s
[CV] END ...................max_depth=2, min_samples_split=2; total time=   0.0s
[CV] END ...................max_depth=2, min_samples_split=2; total time=   0.0s
[CV] END ...................max_depth=2, min_samples_split=3; total time=   0.0s
[CV] END ...................max_depth=2, min_samples_split=3; total time=   0.0s
[CV] END ...................max_depth=2, min_samples_split=3; total time=   0.0s
[CV] END ...................max_depth=2, min_samples_split=3; total time=   0.0s
[CV] END ...................max_depth=2, min_samples_split=3; total time=   0.0s
```

Out[28]:

```
▸ GridSearchCV
▸ best_estimator_: DecisionTreeClassifier
    ▸ DecisionTreeClassifier
```

In [29]:
```
clf=mymodel(clf.best_estimator_)
clf
```

```
Training Score : 0.9666666666666667
Testing score : 0.9333333333333333
Confusion Matric:
            Setosa  Versicolor  Virginica
Setosa          14           0          0
Versicolor       0           7          1
Virginica        0           1          7
Classification report :
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        14
           1       0.88      0.88      0.88         8
           2       0.88      0.88      0.88         8

    accuracy                           0.93        30
   macro avg       0.92      0.92      0.92        30
weighted avg       0.93      0.93      0.93        30
```

Out[29]:

```
▾ DecisionTreeClassifier
DecisionTreeClassifier(max_depth=2)
```

In [30]:

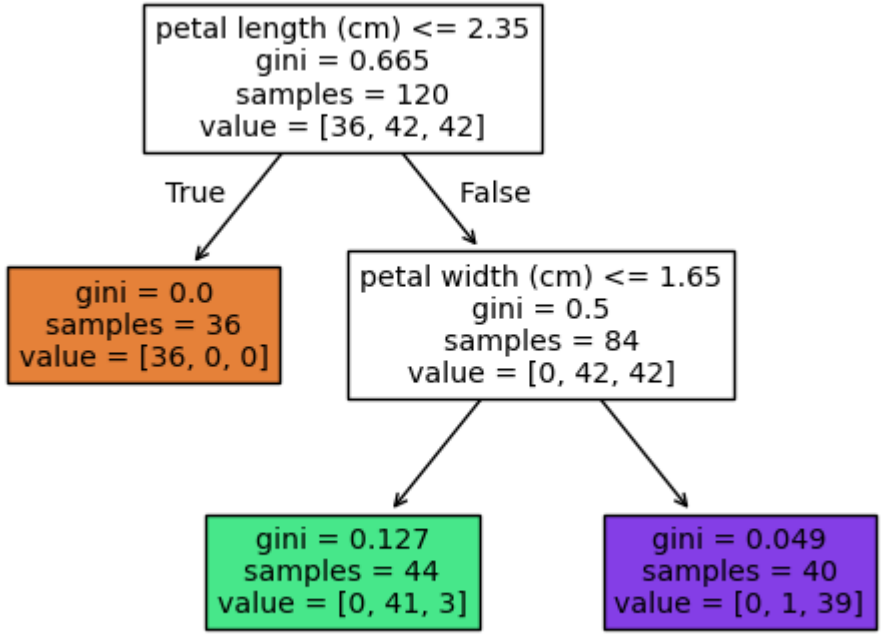
```
f["After pruning Importance"] = clf.feature_importances_
f
```

Out[30]:

| | Features | Importance | After pruning Importance |
|---|---|---|---|
| **0** | sepal length (cm) | 0.000000 | 0.000000 |
| **1** | sepal width (cm) | 0.018797 | 0.000000 |
| **2** | petal length (cm) | 0.056904 | 0.523118 |
| **3** | petal width (cm) | 0.924299 | 0.476882 |

In [31]:

```
from sklearn import tree
tree.plot_tree(clf,feature_names=iris.feature_names,filled = True)
plt.show()
```




pruning techniques and decision trees overfit easily because they keep spliting until they perfectly feet traning data for achiving this tree becomes very complex so the pruning techniques are applied to control the tree complexity. the pre pruning techniques stops the tree before it gross to deep the comon parameters are

1. "max_depth" control maximum depth of the tree.
2. min_sample_split : minimum sample to be present to slipt a node
3. "minimum_sample_leafs" : controls minimum number of samples to be present in a leaf
4. "max_leaf_node": controls maximum number of leaf nodes

GridSerchCV searches fro the best hyper paramaters using cross validation it prevants overfitting try to find the optimal model which is not to complex and use validation folds and not test data. it evaluates model on unseen folds panalies over complex tress and choose the parameters with best generalization.

ensableled techniques combines multiple model to build a strong more stable prediction many weak learnaers gives one strong learner is a key principle types of ensabled techniques : bagging, trains multiple models independently and each model is trained on a sample with replacement also called as bootstrap samples the final prediction is the average or the measurity vote based on regrassion or classification that is why bagging is also called as bootstrap aggrigation its goal is to reduce varince

boosting : boosting here the models are train sequesncially and each model focusses on fixing the previous errors by giving higher weight to the wrong or error done urliear

Random forest is a bagging of dicision tress on randomly selected features each tree gets random rows and random subset of features of each split beacuse of which the trees become less correlated and hence the variance goes down and performance incrase. its key parameters are:

1. "max_depth" control maximum depth of the tree.
2. min_sample_split : minimum sample to be present to slipt a node
3. "minimum_sample_leafs" : controls minimum number of samples to be present in a leaf
4. "max_leaf_node": controls maximum number of leaf nodes
5. also n_estimaters this are the number of trees by default they are 100 also max_fetures this are the maximum number of features to concider number of features

its advantages are :

1. it handles non-linearity
2. it is robhust to noice and gives feature imporatance by giving avergae dicress in impurity

its disadvatages are:

1. it is less explainable and slower

In [35]:
```
import pickle
with open("model.pkl","wb") as f:
    pickle.dump(clf,f)
```

In [ ]:

In [ ]:

In [ ]: